

Lag-1 Autocorrelation in Annual Mean Temperature: A Permutation Test Approach

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1 Introduction

Climatic time series, such as annual mean temperature, often exhibit temporal dependence: temperatures in one year may be related to those in the next. Here, we ask the question:

Are temperatures of one year significantly correlated with the next year (successive years) in a given location?

To answer this, we examine a time series of annual mean temperatures from Key West, Florida, and test whether there is significant lag-1 autocorrelation (i.e. correlation between year t and year $t + 1$).

Because successive measurements in a climatic time series are not independent, we cannot rely on the standard parametric p -value associated with a correlation coefficient. Instead, we use a permutation (randomisation) test to obtain an approximate, non-parametric p -value.

2 Methods

2.1 Data

We used the dataset `KeyWestAnnualMeanTemperature.RData`, which contains an object `ats` with two columns:

- **Year:** calendar year,
- **Temp:** annual mean temperature (in appropriate units, e.g. degrees Fahrenheit or Celsius).

Missing temperature values, if any, were removed prior to analysis.

2.2 Observed lag-1 autocorrelation

Let T_t denote the annual mean temperature in year t , for $t = 1, \dots, n$. We form $(n - 1)$ pairs of successive years:

$$(T_1, T_2), (T_2, T_3), \dots, (T_{n-1}, T_n).$$

We then compute the Pearson correlation coefficient between the vectors

$$X = (T_1, T_2, \dots, T_{n-1}), \quad Y = (T_2, T_3, \dots, T_n),$$

using the `cor` function in R with `method = "pearson"`. This gives the observed lag-1 autocorrelation, denoted r_{obs} .

For the Key West data, we obtained:

$$r_{\text{obs}} = 0.3262.$$

2.3 Permutation test

Because annual temperatures in successive years are not independent, the usual t -test for a correlation coefficient is not appropriate. Instead, we use a permutation test to generate a null distribution for the lag-1 correlation under the hypothesis of no temporal structure.

The steps are:

1. **Observed statistic:** Compute the observed lag-1 correlation r_{obs} as described above.
2. **Permutation of the time series:** Randomly permute the temperature series $\{T_t\}$ to obtain a permuted series $\{T_t^*\}$ of the same length, breaking the temporal order.
3. **Recompute lag-1 correlation:** For each permuted series, compute the lag-1 correlation r^* between $(T_1^*, \dots, T_{n-1}^*)$ and $(T_2^*, \dots, T_n^*)$.
4. **Repeat:** Repeat steps 2–3 a large number of times (here $N = 10,000$ permutations), storing each r^* .
5. **Empirical p -value:** The one-sided p -value for the hypothesis H_1 : “positive autocorrelation” is estimated as

$$p_{\text{one-sided}} \approx \frac{\#\{r^* \geq r_{\text{obs}}\}}{N}.$$

A two-sided p -value can be obtained as

$$p_{\text{two-sided}} \approx \frac{\#\{|r^*| \geq |r_{\text{obs}}|\}}{N}.$$

In R, this was implemented by repeatedly calling `sample(temps, replace = FALSE)` inside a loop, computing the lag-1 correlation for each permuted vector.

3 Results

The observed lag-1 correlation in annual mean temperature was:

$$r_{\text{obs}} = 0.3262.$$

Using $N = 10,000$ random permutations, the empirical p -values were:

- One-sided (testing for positive autocorrelation):

$$p_{\text{one-sided}} \approx 0.00000,$$

i.e. in 10,000 permutations, none produced a correlation as large as r_{obs} .

- Two-sided (testing for non-zero autocorrelation):

$$p_{\text{two-sided}} \approx 0.00110.$$

The figure below shows the null distribution of lag-1 correlations obtained from the permutations, with the observed correlation r_{obs} indicated by a red vertical line.

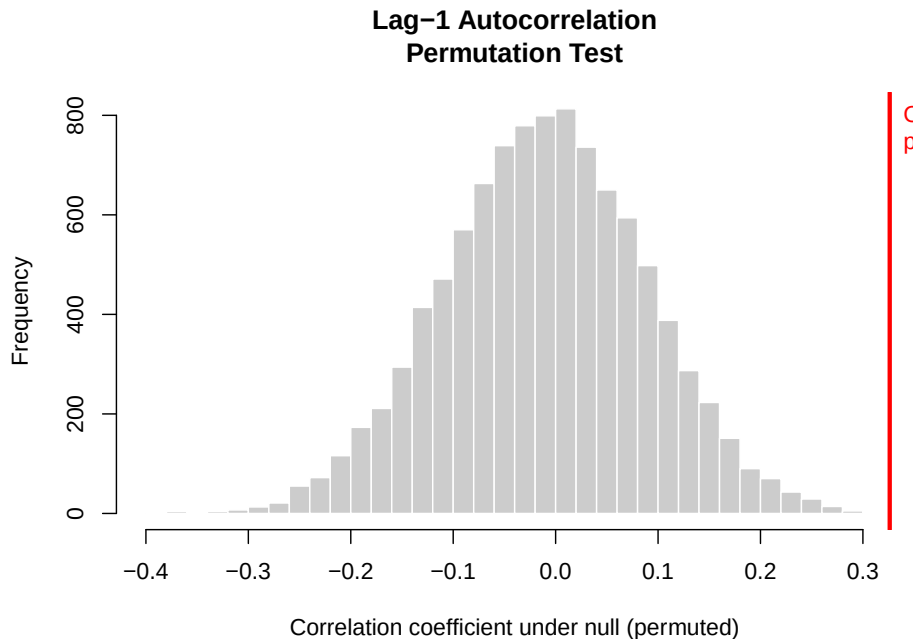


Figure 1: Null distribution of lag-1 correlation coefficients under random permutation of the annual temperature series (grey bars). The red vertical line indicates the observed lag-1 correlation $r_{\text{obs}} = 0.326$.

4 Discussion

The permutation test indicates that the observed positive lag-1 autocorrelation in annual mean temperature is highly unlikely to have arisen by chance under a null model with no temporal dependence. The one-sided empirical p -value is essentially zero ($p \approx 0$), and the two-sided p -value is also very small ($p \approx 0.0011$). Thus, we conclude that annual mean temperatures in this location are significantly and positively correlated between successive years: warmer-than-average years tend to be followed by warmer-than-average years, and cooler-than-average years by cooler-than-average years.

This temporal autocorrelation has important implications for statistical analyses of climatic and ecological responses to temperature. Standard methods that assume independent observations may underestimate uncertainty and inflate type I error rates if they ignore such serial correlation. Permutation-based or time-series methods are therefore preferable when analysing climatic time series.